

O₂RA HEADBAND*

Zsófia Lévai

PHD candidate at Doctoral School

Moholy-Nagy University of Design and Arts, Budapest, Hungary

zsolia.levai@zsolia.com

Abstract

Wearable technology has seen extensive academic exploration but limited commercial diversity, often constrained to devices like smartwatches and rings. This research investigates why broader adoption has stalled, particularly through the lens of materiality and interdisciplinary design. Using a Research through Design (RtD) approach, we developed a soft, textile-based wearable headband integrating a SpO₂ sensor, LED display, and haptic feedback. The project emphasizes the importance of incorporating crafting knowledge—specifically from textile and fashion design—into early prototyping stages. Our findings highlight significant gaps in current wearable development processes, including inadequate methods for integrating rigid and flexible materials, and underutilized expertise in soft material behavior. We argue that rethinking wearable prototyping through collaborative, gendered crafting practices can lead to more functional, comfortable, and innovative designs, with potential applications in fields like neonatal care and beyond.

CCS Concepts

• Human-centered computing-Interaction design • Interaction design process and methods • Interface design prototyping

Keywords

soft interface, crafting, custom made electronics, wearables, textile design expertise

ACM Reference format:

Zsófia Lévai. 2025. O₂RA HEADBAND. In *n Companion of the the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/1234567890>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.

UbiComp Companion '25, October 12–16, 2025, Espoo, Finland

© 2025 Copyright is held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1477-1/25/10.

<https://doi.org/10.1145/3714394.3756226>

1 Introduction

Over the last two decades, our relationship with electronic devices has become increasingly intense and intimate. There is a growing interest in examining these devices more closely and exploring how we might reinterpret and expand them as malleable, tangible user interfaces [1]. This interest has been present since at least the 1990s, as demonstrated by two pioneering publications that already suggested progressive approaches [5][4]. Despite this early promise, commercial wearables today are still largely limited to smartwatches and smart rings. While the field of Human-Computer Interaction (HCI) has seen extensive academic research on wearables, as compiled by Rallabandi et al. [6], there is still the lingering question: why haven't wearables seen broader adoption? The HCI community continues to debate why progress in this area has stalled [3][2].

This research contributes to that ongoing discourse by adopting a Research through Design (RtD) methodology to craft a wearable headband with an integrated SpO₂ sensor. In particular, it explores wearable technology through the lens of gendered crafting practices, seeking to uncover new insights and approaches in wearable design.



Figure 1: Headband with integrated SPO₂ sensor, haptic button, flexible LED screen

2 Problems of wearables from a crafting perspective.

Traditional tangible HCI prototyping typically starts with hardware assembled using overengineered evaluation boards. This is followed by software development customized to the hardware setup. Product design tends to be the final step, often reduced to enclosing the electronics in rigid casings.

To incorporate soft materials and move beyond rigid enclosures, developers usually have two options: collaborate with experts in

textile or material crafts, or self-learn and experiment with those skills. In HCI, the latter—self-training—is more common.

This prototyping hierarchy presents two key limitations:

- When crafting is treated as the final step, it restricts physical affordances and limits the exploration of alternative forms and materials.
- Excluding trained textile or fashion designers risks overlooking valuable material knowledge—potentially compromising both the functionality and evaluability of the final design.

A central challenge in wearable design is flexibility, which encompasses a range of material properties. For instance:

- Soft and bendable but not stretchable (e.g., woven fabrics, conductive threads)
- Bendable but not soft (e.g., flexible PCBs, certain types of leather)
- Soft, bendable, and stretchable (e.g., knitted fabrics)

Each type requires different assembly and construction techniques. For example, sewing woven fabrics differs fundamentally from knitting or embedding electronics in stretchy materials. Successful integration hinges on aligning flexibility profiles—especially when combining soft materials with rigid or semi-rigid components.

The textile and fashion industries have accumulated deep knowledge on how materials interact with the human body, particularly regarding comfort and flexibility. Yet, wearable tech collaborations often lack input from these fields. This underrepresentation risks omitting insights that could significantly improve the wearability and performance of these devices.

3 Research through Design Methodology

This research presents an alternative prototyping approach that incorporates diverse professional expertise from the outset.

The multidisciplinary team that developed the wearable headband included both electrical engineers and textile designers, working in equal collaboration throughout the process.

The development unfolded in two distinct phases:

- Module Integration Phase:

Each module—such as the LED screen, SpO₂ sensor, and haptic button—was individually integrated into soft material layers. These components were tested and stabilized independently before proceeding.

- Wearable Assembly Phase:

Insights from the first phase informed a full redesign. The modules were then combined into a cohesive, three-dimensional soft structure designed to comfortably wrap around the head.

This iterative and collaborative process emphasized material behavior, form, and functionality equally allowing for a more holistic wearable design.

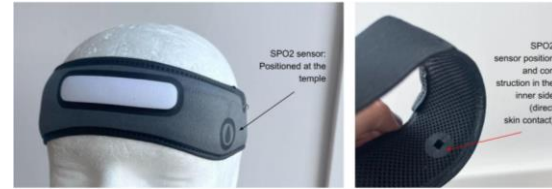


Figure 2: Details of the headband



Figure 3: Details of the battery integration

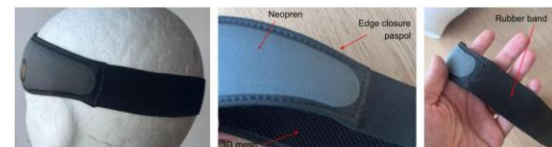


Figure 4: Details of the assembly process

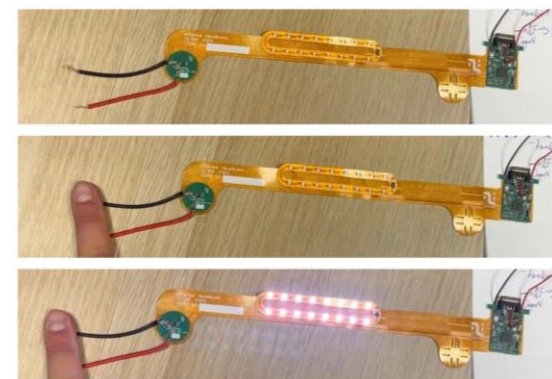


Figure 5: flexible PCB containing LED, main controlling unit and haptic button



Figure 6: Details of the integration

5 Conclusion

This research highlights several key findings and challenges in the development of flexible, textile-based wearable devices:

- The tension between secure electrical connections and signal fidelity remains a central challenge in soft wearables.
- Material layering techniques and flexible PCB integration show promise
- A potential real-world application of this technology is in neonatal care—replacing cumbersome cables in hospital settings with soft, wireless alternatives to monitor vital signs in infants.

By foregrounding material expertise and collaborative design, this work aims to reframe how we approach the prototyping and adoption of wearable technology.

Acknowledgments

This research was supported by the KDP-8-3_PALY-2022_KDP_2021 Kooperatív Doktor Program of the Ministry for Culture and Innovation from the source of the National Research, Development and Innovation Fund. This was a collaboration between Moholy-Nagy University, Zorobe Soft Electronics, ELTE-Soft Nonprofit Kft.

References

- [1] Alberto Boem and Giovanni Maria Troiano. 2019. Non-Rigid HCI: A Review of Deformable Interfaces and Input. In *Proceedings of the 2019 on Designing Interactive Systems Conference*, June 18, 2019. ACM, San Diego CA USA, 885–906. <https://doi.org/10.1145/3322276.3322347>
- [2] Lars Erik Holmquist. 2023. Bits are Cheap, Atoms are Expensive: Critiquing the Turn Towards Tangibility in HCI. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 19, 2023. Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3544549.3582744>
- [3] Lars Erik Holmquist, Oren Zuckerman, Rafael Ballagas, Hiroshi Ishii, Kimiko Ryokai, and Haiyan Zhang. 2019. The Future of Tangible User Interfaces. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*, May 02, 2019. ACM, Glasgow Scotland UK, 1–6. <https://doi.org/10.1145/3290607.3311741>
- [4] Hiroshi Ishii and Brygg Ullmer. 1997. Tangible bits: towards seamless interfaces between people, bits and atoms. In *Proceedings of the ACM SIGCHI Conference on Human factors in computing systems*, March 27, 1997. ACM, Atlanta Georgia USA, 234–241. <https://doi.org/10.1145/258549.258715>
- [5] Maggie Orth, Rehmi Post, and Emily Cooper. 1998. Fabric computing interfaces. In *CHI 98 Conference Summary on Human Factors in Computing Systems*, April 1998. ACM, Los Angeles California USA, 331–332. <https://doi.org/10.1145/286498.286800>
- [6] Vineetha Rallabandi, Alice C Haynes, Courtney N. Reed, and Paul Strohmeier. 2024. Base and Stitch: Evaluating eTextile Interfaces from a Material-Centric View. In *Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction*, February 11, 2024. ACM, Cork Ireland, 1–13. <https://doi.org/10.1145/3623509.3633363>